Evaluating Performance of Different Models in Churn Prediction: A Comprehensive Study

A Comparative Analysis of Machine Learning Models for Churn Prediction

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***Abstract*— One of the main challenges for e-commerce is keeping customers while maintaining customer relationships and subsequently growing the number of customers. Machine learning methods that provide high accuracy of churn prediction can support such businesses in developing suitable retention strategies. In this paper, a dataset from Skogsro Spa has been used as a case study to compare four Machine Learning Algorithms for churn prediction. Results showed that complex algorithms such as Random Forest and Gradient boosting provide higher accuracy in prediction while also analyzing other Machines Learning methods that have been used for churn prediction in this study.**

Keywords— Machine Learning, Churn Prediction, Customer Retention, Performance Evaluation, E-commerce

I. Introduction

The word "churn" is defined by W. Kamakura et al (2005) as "the tendency for customers to defect or cease business with a company". In several marketing studies which investigates customer loyalty by Dick, A.S.; Basu, K (1994), Gefen, D(2002) and Buckinx, W.; Poel,(2005), it has been confirmed that it is more profitable to retain customers than acquiring new ones. Therefore, customer churn prediction Machine learning methods have been studied in a variety of business domains such as telecom, banking and E-commerce.

In this study, a dataset from the years 2013 to 2024 has been provided by Skogsro spa, which is an SME business based in Sandhem, Sweden, that provides in-site Spa services and sells hair and skin care products on its website.

Our focus in this study is to investigate the performance of four Machine learning methods: Logistic regression, Random Forest, Support Vector Machine (SVM), and gradient boosting, to compare their results to define which is more suitable to the problem at hand.

Our approach to churn prediction involved data preparation which includes several activities such as data exploration, understanding and cleaning, followed by feature engineering to create informative features from the dataset which have been defined as follows: Customer ID, purchase analysis, product purchase quantity, order frequency, promotional features, product categorization, product category preferences, recency of purchases, geographic segmentation, churn history integration, and churn label which has been defined as the crucial feature of this study. model building, and evaluation. Data preparation included cleaning (addressing missing values) and feature engineering (creating new features based on purchase history). We employed four machine learning models: Logistic Regression, Random Forest, Gradient Boosting, and Support Vector Machine. Evaluation using metrics such as accuracy, recall, precision, F1 score, and AUC revealed Random Forest as the best-performing model for churn prediction.

The results show that Random Forest and Gradient Boosting are the best-performing models among other chosen models. Such findings can contribute to defining better strategies for customer retention by enabling the E-commerce business to identify customers at risk of leaving and optimize resource allocation while enhancing customer Lifetime Value.

II. Related work

Several studies have investigated various machine learning methods for churn prediction in various industries. On a survey on Machine Learning Methods for churn prediction by L. Geiler, S. Affeldt, and M. Nadif (2022) 11 supervised and semi-supervised machine learning methods have been explored, including Logistic Regression and Support Vector Machine, and found that ensemble approaches like Random Forest, AdaBoost, Gradient Boosting, and XGBoost are effective for churn prediction, with performance influenced by data sampling approaches. Another study Berger and Kompan (2019) proposed a complex model to improve churn prediction accuracy by capturing multiple sets of features representing user interactions with web applications. Additionally, A. Tamaddoni Jahromi et al. (2014) compared different machine learning methods for churn prediction, including Decision Trees, Boosting, and Logistic Regression, with boosting showing higher accuracy.

I. Ullah, B. Raza, and A. Ka (2019) proposed a churn prediction model using several classification algorithms such as Random Forest, Random Tree, and Logistic Regression, with Random Forest exhibiting better accuracy. In a recent study by R. Prabadevi, R. Shalini, and B. R. Kavitha (2023), many machine learning algorithms were compared for predicting customer churn. Logistic Regression performed well for interpreting data, K-Nearest Neighbors provided accurate predictions, and Random Forest was preferred for

feature subset selection. The study also suggested Neural Network learning algorithms as a viable alternative to statistical predictive approaches in customer churn prediction.

The studies by Jahromi, Stakhovych, and Ewing (2014) and Xiahou and Harada (2022) focus on churn prediction in different contexts, emphasizing the use of data mining techniques and customer segmentation to improve accuracy. Jahromi et al. (2014) aim to develop profit-maximizing retention campaigns in B2B settings by modeling customer churn and considering customer heterogeneity.

These researches emphasize the significance of implementing advanced machine learning methods and techniques and data-driven approaches to effectively address churn prediction challenges across diverse industries.

III. Background

The Machine Learning methods that have been used for churn prediction in this study are Logistic regression, Support vector machines (SVM), Gradient Boosting and Random forests.

Logistic Regression, which is a simple algorithm may encounter issues with complex relationships in churn prediction. Advanced algorithms like Gradient Boosting and SVMs offer better flexibility. Random Forest, a powerful classification algorithm, combines multiple decision trees for churn prediction. Hyperparameters such as the number of trees, maximum depth, and splitting criteria are tuned using grid search. Advanced algorithms like Gradient Boosting and SVMs offer better flexibility. The former can effectively capture complex relationships in churn prediction and is capable of handling large datasets. Moreover, a study by Raeisi and Sajedi (2020) aims to predict customer churn using user behavior and online properties. Their study compares multiple machine learning methods and finds that Gradient Boosted Trees outperform others, achieving an accuracy of 86.90%.

While the latter can model non-linear relationships in churn prediction using kernel functions. The study of Xiahou and Harada (2022) proposes a churn prediction model for E-commerce that integrates K-means customer segmentation with Support Vector Machine (SVM) prediction, achieving improved accuracy by identifying core customer segments.

While Logistic Regression offers more simplicity, it has been proven based on research that advanced algorithms like Gradient Boosting, Random Forest, and SVMs provide greater flexibility and effectiveness in capturing complex patterns in churn prediction tasks.

IV. Data exploration

Identifying and obtaining the best data is the most relevant stage in the construction of a predictive model. The quality of the available data determines the accuracy and predictive power of a model. Data exploration involves understanding the contents, characteristics and patterns of the data that aid in further analysis and methodology selection. It involves various tasks such as data understanding, data cleaning, data visualization and feature engineering.

The primary goal of data exploration is to gain a deeper understanding of the dataset. It helps to identify data preprocessing steps needed to clean, transform and reshape the data to make it suitable for analysis. A thorough data exploration may prompt some research questions that can lead to further investigation and experimentation.

A. Data Understanding

Identifying the data that best suits the type of analysis is crucial in extracting relevant features. The process of understanding and interpreting the data is the most difficult and time-consuming task.

Two datasets in CSV format were provided for the project. The former dataset(Order dataset) contained customer order data from a spa ekoshop and webshop that offers massage and makeup services as well as the sale of skin care, hair care, and related products. The latter dataset(Product dataset) contained a priced list of products sold by the shop.

The Order dataset contains information such as customer id, order id, article number, description, date and zip code which are all relevant to the analysis. Each customer possesses a unique customer id that helps to identify a customer. Each purchase made by every customer is registered with a unique order id. The order id gives insights as to the customer who placed the order, the articles purchased and the date of purchase. Every service or product sold by the spa shop is identified with a unique article number. The description describes the name and details of the articles. The date explains when the order was placed by a customer and the zip code gives an approximate idea of the location of the customer. Table (1) illustrates the summary of data contained in the order dataset.

Table 1 Summary of Order dataset

Column name	***Datatype***	***Unique Count***
Order ID	Integer	57044
Customer ID	Integer	14257
Article Number	Float	3537
Description	String	3749
Price	String	1854
Date	String	1770
Zip code	Float	5178

The product dataset contains a list of articles available for purchase with their article number, a detailed description of the article and the price of each. Table (2) illustrates the summary of data contained in the product dataset.

Table 2 Summary of Product dataset

Column name	***Datatype***	***Unique Count***
Article Number	Integer	1670
Description	String	1670
Price	String	176

It was clear that the Order dataset contains purchase information of 14257 unique customers ranging from 2013 to 2024. It also gives some idea about the purchase pattern of customers like the products or services bought, frequency of purchases, the interval between two purchases and customer spending value. The zip code points out that the customers are spread all over Sweden. To get a precise understanding of any

underlying patterns in the data, the 'Order dataset' set needed a data cleaning process as the next step.

B. *Data Cleaning*

On examining further, it was found that the datasets contain missing values in many columns. Since the Order dataset contains relevant information about customers, the missing values need to be handled appropriately for a better churn prediction. Figure 1 depicts the percentage of missing values present in the dataset.

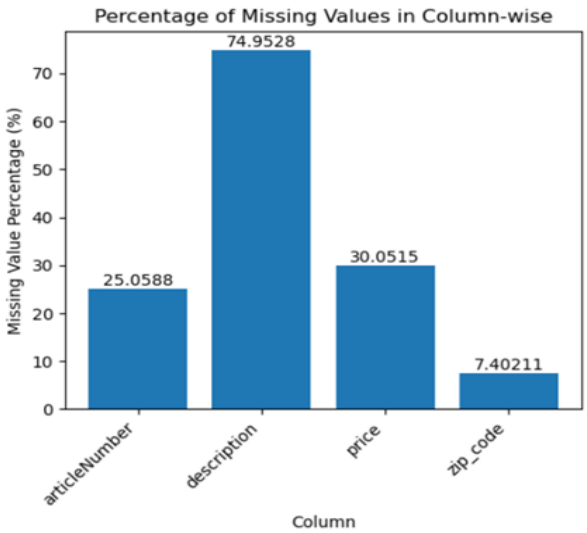

Figure 1: Bar graph showing the percentage of missing values.

To handle the missing values in the 'Description' and 'Price' columns, both datasets were merged based on matching Article Number. As a result, the missing prices and descriptions were imputed to a large extent. The missing values in the 'Description' column were reduced from 74.95% to 15.90% and the 'Price' column was reduced to 16.63%. Since the customer didn't need to register a zip code while making purchases from the store in person, the missing values in 'zip code' represented 'Instore purchase'. Hence the missing values in 'zip code' were replaced by a new label 'Instore purchase'.

But the missing values in 'Article Number' posed a specific challenge. To address this, both datasets were again merged based on matching 'Description'. This method could fill up only 11 instances in 'Article Number'. Since the dataset contained descriptions corresponding to the missing values in 'Article Number' a categorization of products was performed which grouped all similar entities (discounts, samples, gifts, massage, face care products, body care products, hair care products, etc). After this, a unique article number was assigned to each category that filled up all the missing values in 'Article Number'.

To fill the missing values in 'Price' was the next challenging task. It was found from the description that some were samples that were offered for free and many of them lacked description which made it impossible to identify the product type. Hence the appropriate method to fill the missing values in prices would be to give 0 value to avoid any biases in customer spending.

Removing meaningless or irrelevant data from a dataset is crucial for ensuring the accuracy and effectiveness of churn prediction models. The dataset becomes more focused and representative of the actual customer behavior. Some article numbers were found to be meaningless by going through their description and were removed from the dataset. For example, Article number: 99999999 had the description: "Not enough bonus points". Since it doesn't correspond to any particular product or service purchased by a customer, including it in the dataset could introduce noise and inaccuracies into churn prediction models leading to erroneous predictions. 752 such instances were removed from the given dataset.

Another error encountered was in the 'date' column. The original dataset had this column in string format, so typo errors lay undetected, which were corrected and included in the dataset since it contained all relevant information in other feature columns.

Changing data types is often necessary to ensure that the data is in a format that is suitable for analysis or modeling. Converting data to the correct data type ensures that it is stored and processed accurately. For example, if numeric data is stored as text, arithmetic operations may not work correctly. As a last step in the data cleaning process, data types had to be changed to the required format to enable the extraction of certain characteristics and the creation of new features.

C. *Feature Engineering*

Following data cleaning, feature engineering steps were conducted to create additional informative features from the dataset. The initial phase of feature engineering involved grouping the data according to customer ID. This process allowed in summarizing each customer's purchase history, providing valuable insights into their buying behavior. Additional features were created to reveal the buying patterns of customers. These created features fall into the following categories:

- Purchase Analysis: Three features were computed, 'total price spend', 'average price spend', and 'customer spend value' to better understand customer buying patterns. The total amount paid by a customer on all of their orders is represented by their total price spent. On the other hand, the average price spent shows how much a customer typically pays on each order, indicating their regular purchasing patterns. Based on the average spending by the customers an additional feature was created to categorize customers into three segments (High Spending, Medium Spending, and Low spending).
- Product Purchase Quantity: A feature to determine the number of products a customer has purchased. This count excludes non-product purchases like discounts or gift items. To separate actual product purchases, specific product IDs associated with non-product purchases like discounts, coupons, or samples were excluded from the calculation.
- Order Frequency: An indicator of how many orders a consumer has placed overall. This metric shows how frequently a consumer usually makes purchases from the webshop and is a valuable indicator of their buying frequency.

- Promotional Features: Binary features that indicate whether the customer has received any discount, gift, or sample products or not. They can be valuable in understanding the relationship between churn behavior and promotions received.
- Product Categorization: To study more about customer purchase patterns, knowledge about the products purchased by a customer is a very relevant factor. Product categorization plays a crucial role in understanding customer behavior. Categorizing products can help to identify patterns in the types of products that customers usually purchase before churning. It can give better insights into churn prediction like whether customers who churn tend to purchase items from certain categories leading up to churn.
- Product Category Preferences: Binary features that categorize customer purchases based on product types (hair care, body care, face care). This segmentation will help in understanding a customer's priority towards specific product categories. By analyzing churn rates within each segment, the identification of segments that are more prone to churn becomes easy.
- Recency of Purchases: Denotes the latest date for which the customer made the purchase. This feature is used for creating the target feature for the dataset.
- Geographic Segmentation: A feature that denotes the buying patterns across different regions. This was created by categorizing customer zip codes into 14 distinct areas.
- Churn History Integration: To leverage a customer's past churning behavior for improved prediction accuracy, a churn history feature was included. This feature indicates whether a customer has a history of churning from the webshop.
- Churn Label: A crucial aspect of this project was defining customer churn and creating a corresponding label for the dataset. Churn refers to a customer who stops ordering from the spa's ekoshop and webshop. Based on the suggestion of the company, a customer is considered to be churned if the time elapsed since their last purchase exceeded a predefined threshold of 9 months. To determine the time since the last purchase, equation 1 was used. If the time since the last purchase is greater than the threshold the label indicates 'churned customer' otherwise it denotes that the customer is active. If the time since the last purchase is greater than the threshold the label indicates 'churned customer' otherwise it denotes that the customer is active.

$$\text{Equation 1: } Time\ Since\ Last\ Purchase = Latest\ Date\ in\ the\ Dataset - Last\ Purchase\ Date.$$

By applying these feature engineering techniques, a comprehensive set of features were created that can be used to understand customer behavior, buying patterns and their preferences. It enhances the predictive models to predict churn with greater accuracy.

D. Data Analysis

Data analysis is the process of inspecting and modeling data to discover useful information and draw conclusions. It involves applying various statistical and computational techniques to interpret patterns, trends, and relationships within datasets. Data analysis can be conducted using many methods. Descriptive statistics and exploratory data analysis (EDA) were employed in this project to analyze the data.

From the figure 2, it can be inferred that there is a lot of variability in the data. Some customers spend a lot of money and place many orders, while others spend very little and place few orders. This could be due to several factors, such as customer demographics, product preferences, or marketing campaigns. One such interpretation from figure 2 can be that 75% of customers have spent less than 1399 sek with an average spending of less than 169 sek per purchase and a maximum purchase of 7 products per order. Around 50% of the customers are single-order customers.

	Total Price Spend	Avg Price Spend	Number Of Products Purchased	Number Of Total Orders Placed
count	14257.000000	14257.000000	14257.000000	14257.000000
mean	2193.517262	127.979854	11.072666	4.032195
std	5609.037361	178.161771	28.579090	8.317663
min	-2985.000000	-1179.000000	0.000000	1.000000
25%	78.000000	29.000000	1.000000	1.000000
50%	449.000000	103.640000	2.000000	1.000000
75%	1399.000000	169.000000	7.000000	3.000000
max	150552.350000	4890.000000	917.000000	317.000000

Figure 2: descriptive statistics of numerical values in the dataset.

Class imbalance is a notable factor in this dataset; as churned customers make up 77.9% compared to only 22.1% of existing customers (Figure 3). This imbalance can bias machine learning algorithms towards the majority class. A technique used to address the imbalance is explained in section V of the report.

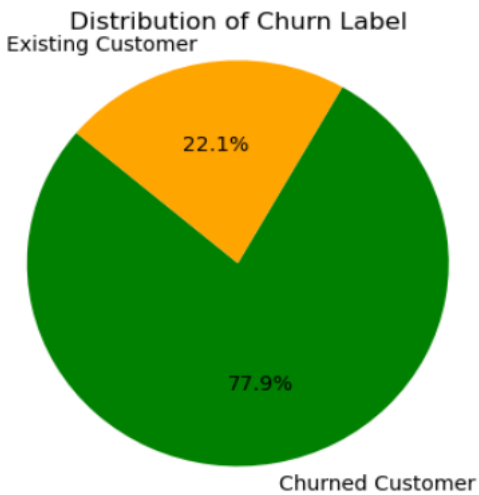

Figure 3: Pie-chart for distribution of the churn- label.

Analyzing customers by average spending often reveals a clear connection between spending habits and churn rate, from figure 4 it can be seen that the churn rate is highest among low-spending customers and lowest among high-spending customers.

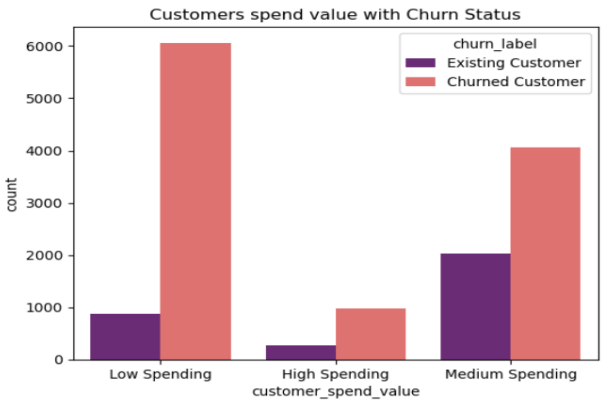

Figure 4: Bar-chart of churn based on customer spending.

Figure (5) shows that the churn rate is minimum among customers who bought hair care products and is maximum among customers who haven't purchased any hair care products. It can also be understood that customer churn is not explicitly dependent on the product purchase alone. Other factors like customer spending, and discounts given may all be responsible factors for churning which is discussed in the feature importance section.

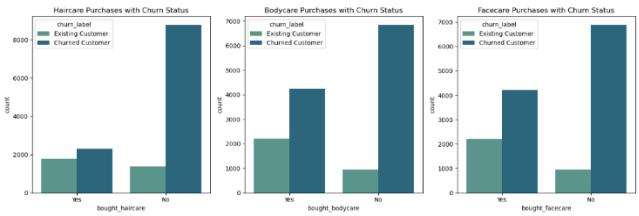

Figure 5: Bar chart illustrating the customer churn based on product purchased.

V. METHOD DESCRIPTION

The following chapter contains different methods used throughout the project to reach the final implementations and conclusions. The subchapters are the different processes used during for final implementations.

A. Tools Used

The project used Jupyter Notebook, Visual Studio Code, and Excel sheets. Jupyter Notebook allows for creating documents with code, math, visuals, and explanations. Visual Studio Code helped with coding tasks, providing a flexible environment. Excel was used for organizing and analyzing data. These tools made project management and analysis easier.

B. Libraries Used

The project used various libraries to streamline the data analysis and modelling processes. These libraries played a crucial role in simplifying the development of a robust program capable of handling large datasets, making predictions, and extracting meaningful insights. Among the essential libraries employed were: .

- Pandas: A Python library that makes working with structured data easy. It offers data structures like DataFrame and Series, which are like tables with rows and columns. Pandas help with tasks such as cleaning, transforming, and exploring data, which are important for preparing data before machine learning.
- ·NumPy: A Python library for numerical computing, emphasizing large arrays and matrices. It includes math functions for efficient array operations. NumPy serves as a base for other libraries like Pandas and scikit-learn.
- Scikit Learn: Scikit-learn, also known as sklearn, is a Python library for machine learning. It offers various algorithms for tasks like classification, regression, and clustering. Scikit-learn also provides tools for data preparation, model assessment, and tuning parameters.
- Matplotlib: A plotting library in Python widely used for creating visualizations, particularly for data exploration and presentation. It offers a wide range of functionalities for creating various types of plots, including line plots, scatter plots, bar plots, histograms, and more.
- Seaborn: A Python data visualization library built on matplotlib. It simplifies creating attractive statistical graphics with minimal code, making it popular for data exploration and presentation.
- Datetime: A Module in Python's standard library used for working with dates and times. They provide functionalities for creating, manipulating, and formatting date and time objects, as well as performing arithmetic operations on them. The Datetime library was used in the calculation of the churn label in this project.

C. Train-Test Split

The dataset was split into training and testing sets using a train-test split method in sklearn. This technique is crucial for evaluating the model's ability to generalize on unseen data. 80% of the data is used for training, allowing the model to learn patterns. The remaining 20% serves as the testing set, unseen by the model during training. This helps assess the model's performance on real-world, unseen data.

D. Standardization

Standardization, a preprocessing step, was implemented to improve the performance of the logistic regression model. It transforms the features in both the training and testing sets to have a mean of zero and a standard deviation of one. This ensures all features contribute equally to the model's predictions. The features in the dataset belonged to different scales, standardization prevents features with larger scales from dominating the model's learning process and faster learning.

E. Addressing the class imbalance

Due to class imbalance in the dataset, where churned customers are outnumbered by existing customers. This can lead to biased models. To address this, SMOTE (Synthetic Minority Oversampling Technique) was employed. SMOTE is an oversampling technique that creates synthetic data points for the minority class (existing customers). This helps balance the class distribution and allows the model to learn more effectively from the minority class, ultimately leading to more accurate churn predictions.

F. *Stratified K-Fold cross-validation*

Stratified K-Fold cross-validation keeps the class proportions same in each fold. It divides the data into 10 folds, using 9 for training and 1 for testing. GridSearchCV uses this method to adjust hyperparameters, testing combinations through cross-validation.

G. *Machine Learning Models and Hyper Parameter Tuning.*

Four different machine learning algorithms were used in model development: Logistic Regression, Random Forest, Gradient Boosting, and Support Vector Machine. In the context of model development, Logistic Regression often serves as a baseline model due to its simplicity and interpretability, similar to Linear Regression. However, more complex algorithms like Gradient Boosting and Support Vector Machines (SVM) offer greater flexibility and can capture non-linear relationships in the data more effectively. Through hyperparameter tuning, the model can improve its performance to contribute effectively to our churn prediction task.

Table 3 displays unique hyperparameters for each algorithm. Adjusting these hyperparameters is vital for optimizing each algorithm's performance and achieving desired results.

Table 3 Hyperparameters used in all models

Table Column Head		
Model	***Parameter***	***Variable Used***
Logistic Regression	Regularization strength(C)	0.001, 0.01, 0.1, 1, 10, 100
Random Forest	n_estimators max_depth min_samples_split min_samples_leaf	[50,200,300] [2, 4,6] [2,3,5] [1, 2]
Gradient Boosting	learning rate number of estimators maximum depth of trees	[0.01, 0.1, 1] [50, 100, 200] [3, 4, 5]
Support Vector Machine	regularization parameter (C) kernel type gamma parameter	[0.1, 1, 10] [linear, rbf, poly, sigmoid] [scale, auto]

Logistic Regression: Logistic Regression models the relationship between independent variables and the probability of a binary outcome. While it's simple and interpretable, it may struggle with complex, non-linear relationships. However, it serves as a foundational model in this project, providing insights into predictor variable significance and serving as a benchmark for comparison with more complex algorithms.

1) Random Forest: Random forest works on the divide and conquer approach. It is based on the random subspace method. In this method, a number of trees are formed and each decision tree is trained by selecting any random sample of attributes from the predictor attributes set. Each tree matures up to a maximum extent based on the attributes or parameters present. The final decision tree formed for the prediction is mainly based on weighted averages.

2) Gradient boosting: Gradient Boosting is a machine learning technique that builds a predictive model by combining several weak learners sequentially. Each weak learner is trained to correct the errors made by the previous ones. In essence, Gradient Boosting works by iteratively improving the model's predictions. It does this by emphasizing the data points that the previous models misclassified, thereby gradually reducing prediction errors. This iterative process continues until the model achieves a certain level of accuracy or until a predefined number of weak learners have been added.

3) Support Vector Machine: The Support Vector Machine (SVM) algorithm works by finding the optimal hyperplane that best separates data points belonging to different classes in a high-dimensional space. It does this by maximizing the margin, which is the distance between the hyperplane and the nearest data points from each class, known as support vectors. SVM can handle linearly separable data by finding a hyperplane directly, but for non-linearly separable data, it uses a kernel trick to map the data into a higher-dimensional space where it can be linearly separated. This allows SVM to classify data points even when they are not linearly separable in the original space.

VI. EVALUATION

Evaluation in machine learning is the process of assessing a model's performance on unseen data. It helps determine the model's predictive accuracy and generalization ability. By comparing different models or variations, evaluation aids in selecting the best-performing solution for a given task. Additionally, it provides insights into model strengths and weaknesses, guiding improvements and informing decision-making processes.

As a result, we assess the performance of four machine learning models - Logistic Regression, Support Vector Machine (SVM), Random Forest, and Gradient Boosting - in predicting customer churn for Skogsro Spa's webshop based on past order data. The evaluation is conducted using metrics such as accuracy, recall, precision, F1 score, and area under the ROC curve (AUC). Furthermore, we analyze the confusion matrix for the Random Forest model to gain insights into its predictive capabilities.

A. *Comparison of Evaluation Results*

Table 4 compares the performance of the four models trained based on different evaluation metrics:

Model	**Accuracy**	**Recall**	**Precision**	**F1 Score**
Gradient Boosting	0.8015	0.8521	0.8917	0.8714
Logistic Regression	0.7703	0.7948	0.9026	0.8453
SVM	0.7563	0.7766	0.9010	0.8342
Random Forst	0.7563	0.9684	0.8542	0.9078

Table 4: Comparison of 4 models.

- Accuracy: Accuracy indicates the overall correctness of the model's predictions. Among the models, Random Forest achieves the highest accuracy of

84.47%, followed by Gradient Boosting (80.15%), Logistic Regression (77.03%), and SVM (75.63%).

- Recall: Recall measures the proportion of actual churners that are correctly identified by the model. Random Forest outperforms other models in recall with a value of 96.85%, indicating its effectiveness in capturing true positives. Gradient Boosting follows with a recall of 85.21%, while Logistic Regression and SVM achieve recall values of 79.49% and 77.66% respectively.
- Precision: Precision quantifies the accuracy of positive predictions made by the model. Here, Gradient Boosting demonstrates the highest precision of 89.17%, followed closely by Logistic Regression (90.27%) and Random Forest (85.43%). SVM lags slightly behind with a precision of 90.11%.
- F1 Score: The F1 score is the harmonic mean of precision and recall, providing a balanced assessment of a model's performance. Random Forest exhibits the highest F1 score of 90.78%, followed by Gradient Boosting (87.15%), Logistic Regression (84.53%), and SVM (83.42%).

Based on these metrics, Random Forest emerges as the best-performing model overall, with superior accuracy, recall, precision, and F1 score compared to other models. However, it's important to note that Gradient Boosting also shows competitive performance across multiple metrics, indicating its potential effectiveness in predicting customer churn for Skogsro Spa.

B. *Receiver Operating Characteristic (ROC)*

Figure 6 is the illustration of the ROC of each classification model. The ROC curve analysis offers valuable insights into the predictive performance of machine learning models for customer churn prediction at Skogsro Spa's webshop. By evaluating the trade-off between sensitivity (true positive rate) and specificity (true negative rate) at various classification thresholds, the ROC curve provides a comprehensive understanding of each model's discriminatory power.

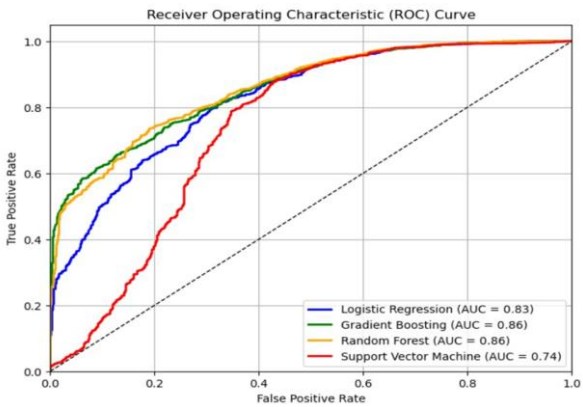

Figure 6: ROC Curve for all the models

Random Forest and Gradient Boosting exhibit the highest Area Under the Curve (AUC) values, both scoring an impressive 0.86, indicating strong discriminatory ability. Logistic Regression also demonstrates good discriminatory power with an AUC of 0.83, while Support Vector Machine (SVM) lags with an AUC of 0.74. These results guide Skogsro Spa in selecting the most effective model for targeting marketing efforts and optimizing customer retention strategies. Models with higher AUC values are better suited for identifying potential churners, enabling more targeted and personalized marketing campaigns. Additionally, understanding false positive and false negative rates allows for the refinement of customer segmentation strategies, optimizing resource allocation for retention efforts. Regular monitoring of ROC curves and AUC values enables Skogsro Spa to assess model performance over time and identify potential drift or degradation in predictive accuracy. This iterative approach fosters continuous improvement, ensuring that predictive models remain effective in dynamic market conditions. By leveraging predictive analytics effectively and continuously monitoring model performance, the spa can stay ahead in the competitive spa and skincare market, maximizing customer satisfaction and business success.

C. *Confusion Matrix*

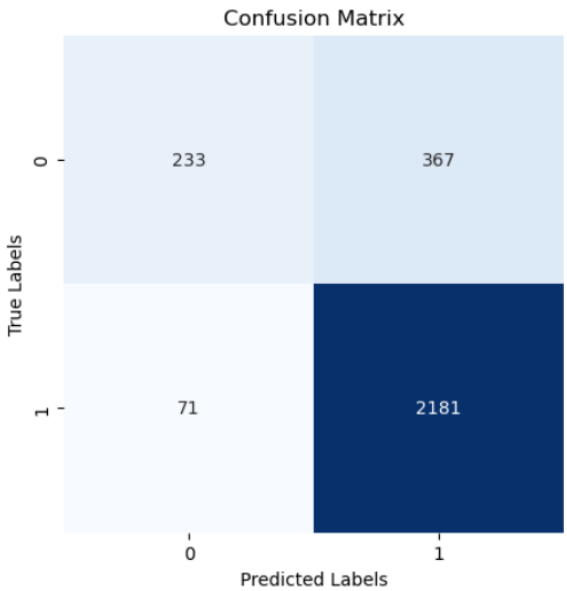

Figure 7: Confusion Matrix

Figure 7 is the confusion matrix analysis provides detailed insights into the predictive performance of the Random Forest model for customer churn prediction. Among the four quadrants of the matrix, the Random Forest model correctly identified 2181 customers who were predicted to churn and indeed churned, representing true positives (TP). Additionally, the model accurately classified 233 customers as loyal, who continued ordering from the webshop as predicted, denoted as true negatives (TN). However, the model also misclassified 367 customers as likely to churn when they continued their purchases (false positives, FP), which may lead to misallocation of retention resources. Furthermore, the model failed to identify 71 customers who churned, classifying them as likely to continue ordering (false negatives, FN), which indicates the missed opportunities for intervention and retention.

D. *Feature Importance Analysis*

To understand the factors influencing churn, a feature importance analysis was conducted on the Random Forest model. This analysis shows the relative contribution of each feature to the model's prediction as illustrated in Figure 8. The analysis highlights that the Total Price Spend and Number of Products Purchased are the most significant churn indicators which indicate that the Customers with higher spending and purchase frequency are less likely to churn. It could be also analysed that the average price Spend has a lower importance compared to total spending, suggesting a focus on overall spending habits rather than the individual purchase value. From product-level analysis, the customers who bought hair care products are less likely to churn when compared to customers who bought face care and body care products. All other features contribute less to the churn when compared.

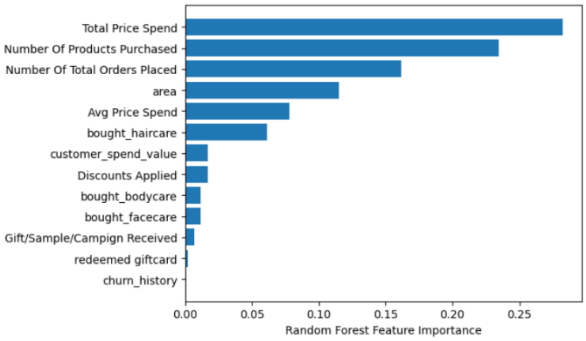

Figure 8: Bar Chart depicting the feature importance.

VII. Conclusion

Overall, assessing predictive modelling methods for anticipating customer churn at Skogsro Spa's online store offers important findings on how machine learning algorithms can improve customer retention strategies and optimize business results. By examining performance metrics like accuracy, recall, precision, F1 score, AUC, and the confusion matrix in depth, we have compared the performance of Logistic Regression, Support Vector Machine (SVM), Random Forest, and Gradient Boosting. The findings indicate that Random Forest and Gradient Boosting are the top-performing models. This report also pinpoints the strengths and potential areas of enhancement in the best-performing model Random Forest.

Analyzing the feature importance of the Random Forest model reveals some interesting insights into customer churn behavior. By using the insights from the Random Forest model, Skogsro Spa can improve marketing campaigns to encourage higher purchase frequency and total order values. Implementing loyalty programs and personalized recommendations based on past purchase behavior can further strengthen customer relationships. Additionally, exploring regional trends in churn can help refine marketing strategies for specific locations. While promotions may play a role in customer acquisition, the data shows that these customers have made only a single purchase. Hence, focusing on building long-term value and satisfaction appears to be more crucial for customer retention.

Overall, this project allowed us to gain valuable and practical experience in implementing machine learning algorithms for churn prediction. We learned the importance of fine-tuning the models and feature engineering to achieve better results. It is important to note that the prediction accuracy could be further improved by exploring a wider range of more complex models which was not possible in this study due to time limitations. Future research could also investigate the inclusion of additional data related to customers such as age, and gender to enhance the predictive power of the model. Furthermore, external data regarding the presence of business competitors that play a role in customer churn could be included for better prediction.

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